

PAPER CODE: BCE-P614
SEMESTER EXAMINATION (MAY-2025)
CLASS: B. TECH SEMESTER: VI
SUBJECT: CSE
PAPER TITLE: MACHINE LEARNING-II

Time: 3 Hours

Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)		CO	BL
Instructions: Answer any <i>five</i> questions in about 150 words each. Each question carries six marks. (5 x 6 = 30 Marks)			
Q-1	What is the main objective of a regression problem in machine learning?	CO2	B2
Q-2	Differentiate between overfitting and underfitting with examples.	CO1	B2
Q-3	What is the basic idea behind Bagging and Boosting in ensemble learning?	CO4	B2
Q-4	What is polynomial regression and how does it differ from linear regression?	CO2	B2
Q-5	Write a Top - Down Approach Constructing a Decision Tree.	CO3	B3
Q-6	What is the main objective of a Support Vector Machine (SVM) algorithm in classification problems?	CO2	B2
Q-7	How KNN Works?	CO2	B2
Q-8	What is the Gradient Descent Method? How Does Gradient Descent Work in Linear Regression?	CO2	B3
Q-9	What is K-Means clustering and how does it work?	CO5	B3
Q-10	How does NumPy differ from Pandas in terms of data manipulation?	CO4	B1
SECTION-B (LONG ANSWER TYPE QUESTIONS)		CO	BL
Instructions: Answer any <i>four</i> questions in detail. Each question carries 10 marks. (4 x 10 = Marks)			
Q-1	Explain distance metrics for classification with examples?	CO1	B2
Q-2	Suppose we have a dataset of news articles with their content tokenized into sets of keywords. Article-: {1,2,3,4,5}. Keywords (set of words)-: {{politics, economy, market}, {Stock, economy, market}, {Government, Tax, Budget}, {Music, Album, Tour}, {Bond, Finance, Investment}}. Category-: {Politics, Business, Politics, Entertainment, Business} Consider a new article with the keywords {Economy, Stock, Bond} using K-nn Classifier for k=3. Use Jaccard index as the distance metric.	CO5	B4

Q-3

Candidate	Skill Level	Certification	Previous Job Success	Education Relevance	Hired
1	High	Yes	High	Relevant	Yes
2	High	No	Low	Irrelevant	No
3	Medium	Yes	Medium	Relevant	Yes
4	Low	No	Low	Irrelevant	No
5	Medium	No	Medium	Irrelevant	No
6	Low	Yes	High	Relevant	No
7	High	Yes	Medium	Irrelevant	Yes
8	Medium	Yes	Low	Relevant	No
9	High	No	High	Relevant	Yes
10	Low	No	Medium	Irrelevant	No

Classify whether a new candidate with skill level: Medium, has done certification, Previous Job Success: High, Education Relevant as hired or not Using Navie Bayes?

CO3 B4

Q-4

Consider a small dataset Containing information about weather conditions and whether to play Tennis or not. Construct a Decision Tree using Top-Down Approach using Information Gain as the Splitting Criteria.

CO3 B4

Outlook	Temperature	Humidity	Windy	Play Tennis
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rain	Mild	High	False	Yes
Rain	Cool	Normal	False	Yes
Rain	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rain	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rain	Mild	High	True	No

Q-5	What is the Random Forest? List and explain the advantages and Disadvantages of the Random Forest algorithm.	CO4	B2																					
Q-6	What are Clustering and Types of Clustering Methods?	CO4	B2																					
Q-7	Consider the following 2D dataset with six data points. <table><tr><th>Data Point</th><th>Feature 1. (x1)</th><th>Feature 2. (x2)</th></tr><tr><td>1</td><td>1.0</td><td>2.0</td></tr><tr><td>2</td><td>1.5</td><td>1.8</td></tr><tr><td>3</td><td>5.0</td><td>8.0</td></tr><tr><td>4</td><td>8.0</td><td>8.0</td></tr><tr><td>5</td><td>0.6</td><td>0.6</td></tr><tr><td>6</td><td>11.0</td><td>11.0</td></tr></table> Let's choose $k=2$ and apply the k-means algorithm with initially chosen centroids as data point 1 and 3.	Data Point	Feature 1. (x1)	Feature 2. (x2)	1	1.0	2.0	2	1.5	1.8	3	5.0	8.0	4	8.0	8.0	5	0.6	0.6	6	11.0	11.0	CO5	B3
Data Point	Feature 1. (x1)	Feature 2. (x2)																						
1	1.0	2.0																						
2	1.5	1.8																						
3	5.0	8.0																						
4	8.0	8.0																						
5	0.6	0.6																						
6	11.0	11.0																						
Q-8	Write a Python program that computes the parameters of a linear regression model using the Least Squares Method and visualizes the best-fit line using Matplotlib.	CO6	B5																					
